GREEEN tallgrass prairie restoration: Seeded richness ~ Species pool size (SW Station)

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> **NOTE:** To compare different models, you only have to change the models in the section 'Load models'

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```r
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
library(lme4)
library(glmmTMB)
library(DHARMa)
library(emmeans)
```

Packages

```r
sites <- read_csv(
  here("data", "processed", "data_processed_sites.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    id_plot = "f",
    year = "f",
    seeding_time = col_factor(
      levels = c("unseeded", "fall", "spring"), ordered = FALSE
      ),
    seeded_pool = col_factor(
      levels = c("0", "6", "12", "18", "33"), ordered = TRUE
      ),
    treatment_id = "f",
    richness_type = "f"
  )
) %>%
  filter(
    site == "SW Station",
    richness_type == "seeded_richness",
    !(treatment_id %in% c("2", "4"))
  ) %>%
  mutate(y = richness_1qm + richness_25qm)
```

Load data

Statistics

Data exploration

Means and deviations

```r
Rmisc::CI(sites$y, ci = .95)
```

```
##    upper     mean    lower
## 3.287302 2.984375 2.681448
```

```r
median(sites$y)
```

```
## [1] 3
```

```r
sd(sites$y)
```

```
## [1] 2.128039
```

```r
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##  5% 95%
##   0   7
```

```r
sites %>% count(site, year)
```

```
## # A tibble: 4 x 3
##   site       year      n
##   <chr>      <fct> <int>
## 1 SW Station 2015     48
## 2 SW Station 2016     48
## 3 SW Station 2017     48
## 4 SW Station 2018     48
```

```r
sites %>% count(seeded_pool)
```

```
## # A tibble: 4 x 2
##   seeded_pool     n
##   <ord>       <int>
## 1 6              48
## 2 12             48
## 3 18             48
## 4 33             48
```

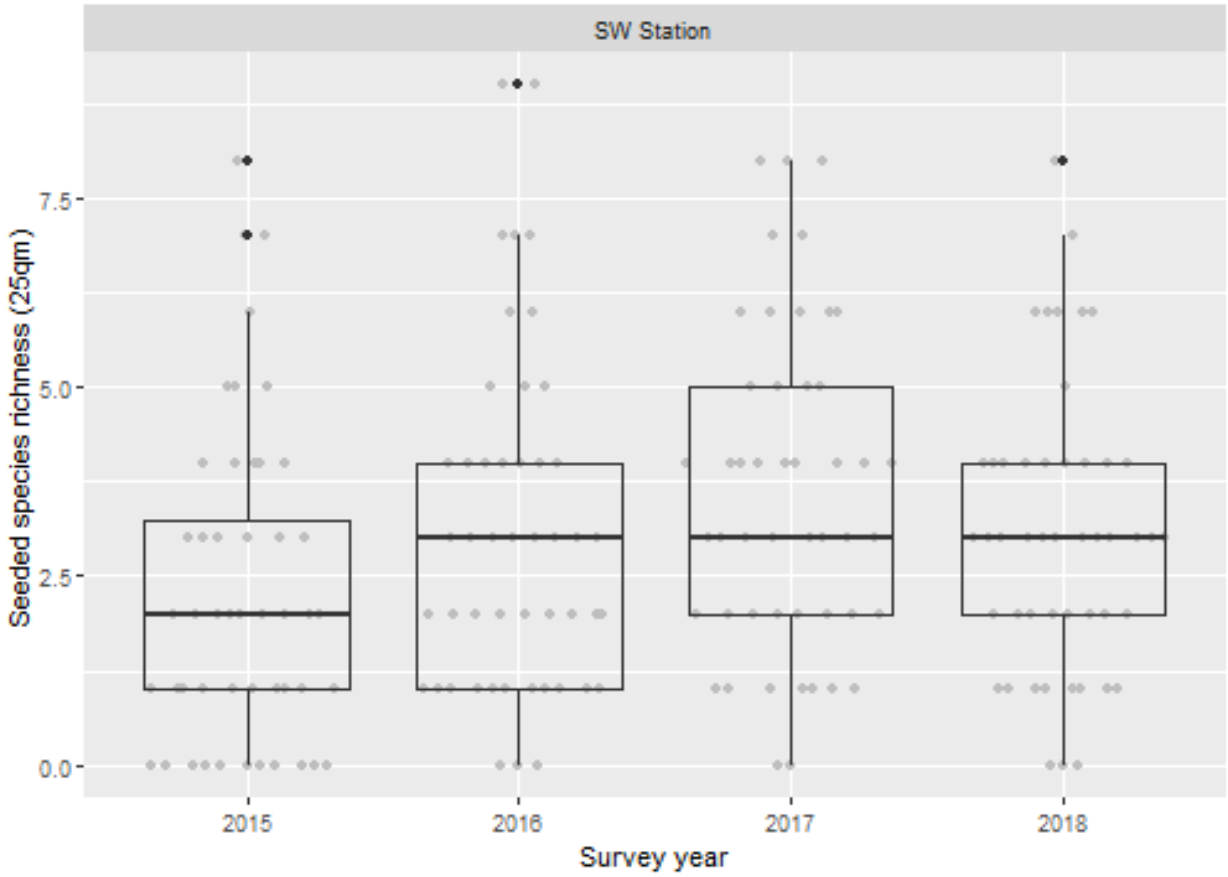
```r
sites %>% count(seeding_time)
```

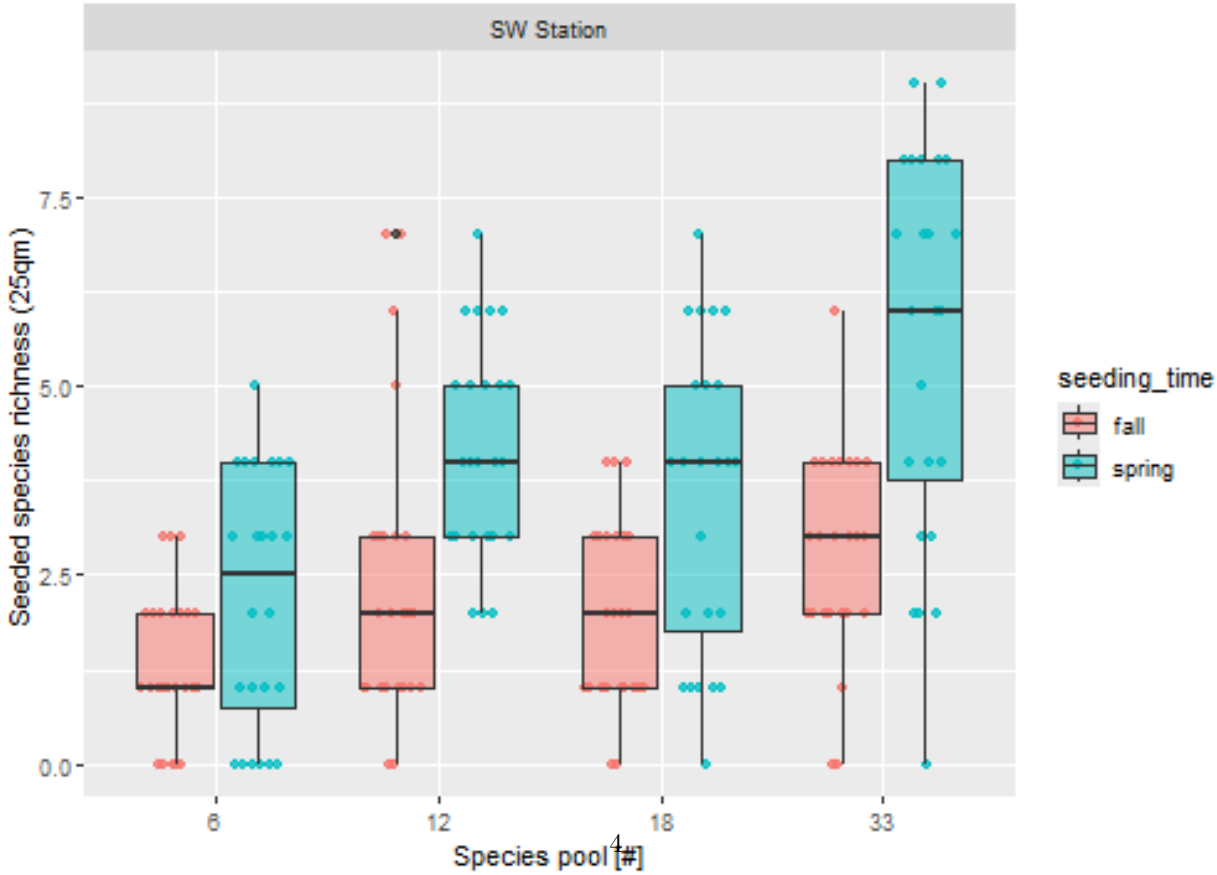
```
## # A tibble: 2 x 2
##   seeding_time     n
##   <fct>        <int>
## 1 fall            96
## 2 spring          96
```

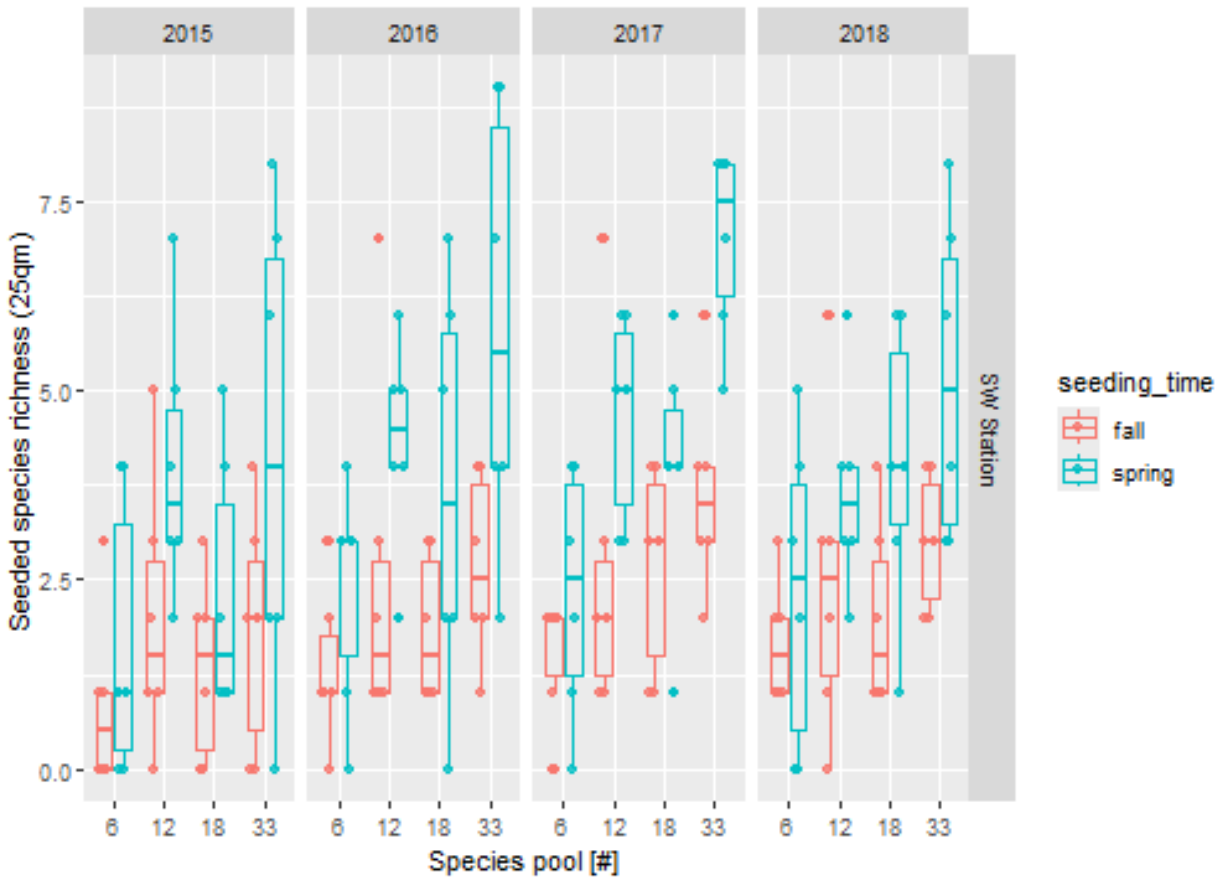
```r
sites %>% count(herbicide)
```

```
## # A tibble: 2 x 2
##   herbicide     n
##       <dbl> <int>
## 1         0    96
## 2         1    96
```

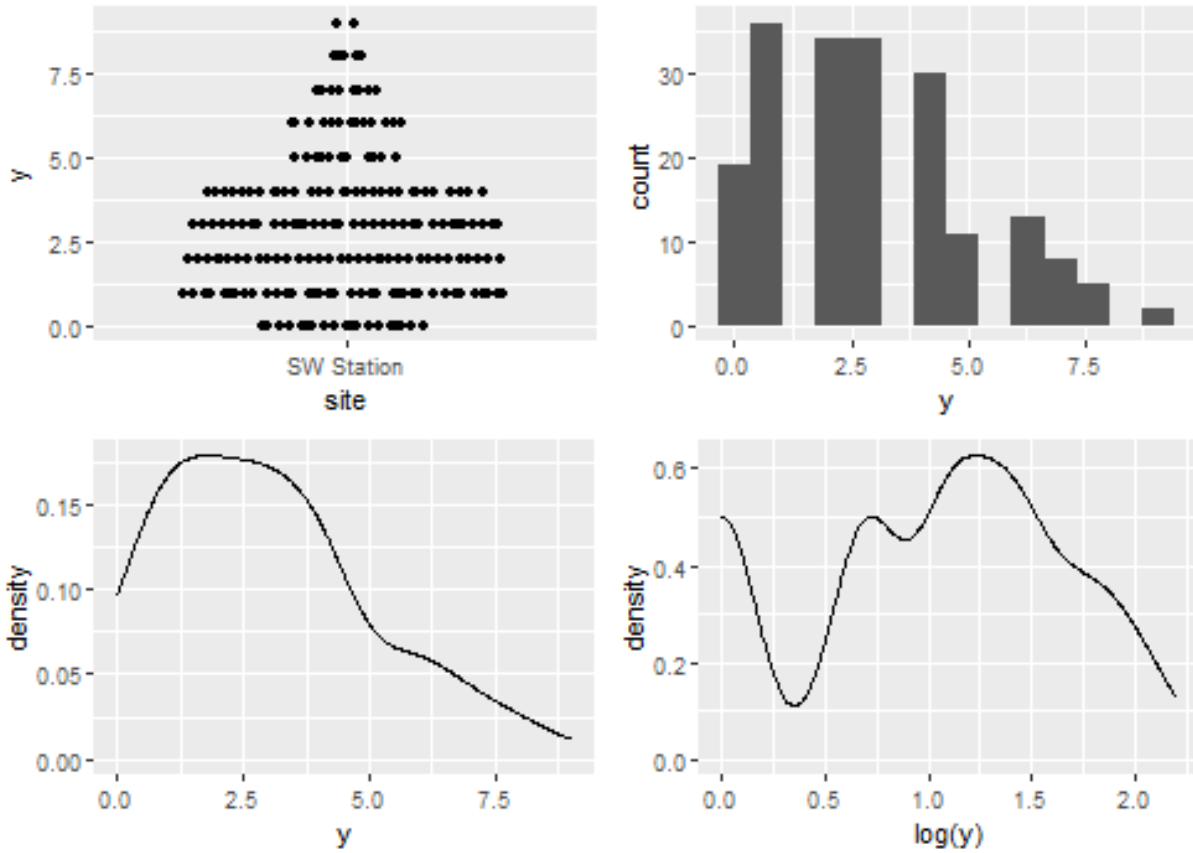
Graphs of raw data (Step 2, 6, 7)






2015
2016
2017
2018
SW Station
Seeded species richness (25qm)
7.5
5.0
2.5
0.0
6
12
18
33
Species pool [#]
seeding_time
fall
spring

Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude r > 0.7 Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cwm.")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#     ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_richness_pool_sws_2.Rdata"))
load(file = here("outputs", "models", "model_richness_pool_sws_3.Rdata"))
m_1 <- m2
m_2 <- m3
```

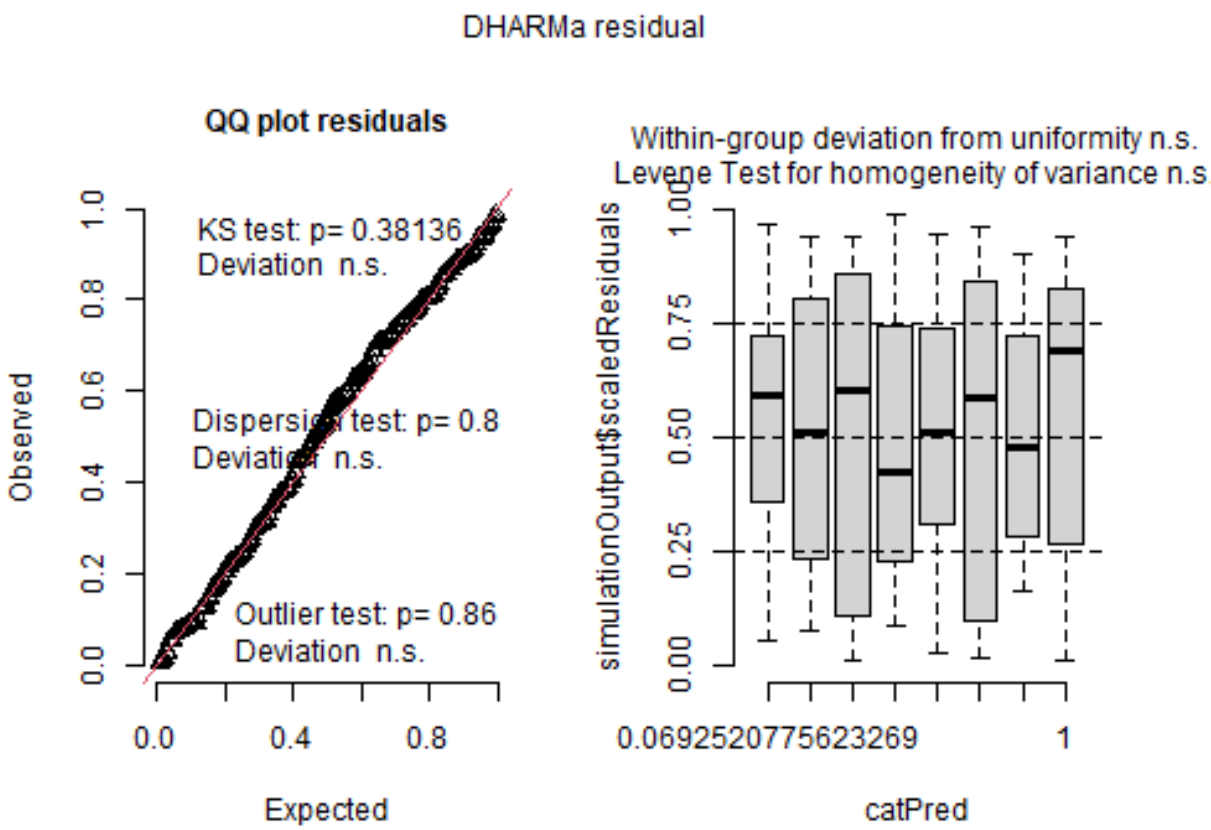
```
# formula(m_1)
# formula(m_2)
m_1@call
```

```
## glmer(formula = y ~ seeded_pool + seeding_time + (1 | year),
##     data = sites, family = poisson(link = "log"))
m_2@call
## glmer(formula = y ~ seeded_pool * seeding_time + (1 | year),
##     data = sites, family = poisson(link = "log"))
```

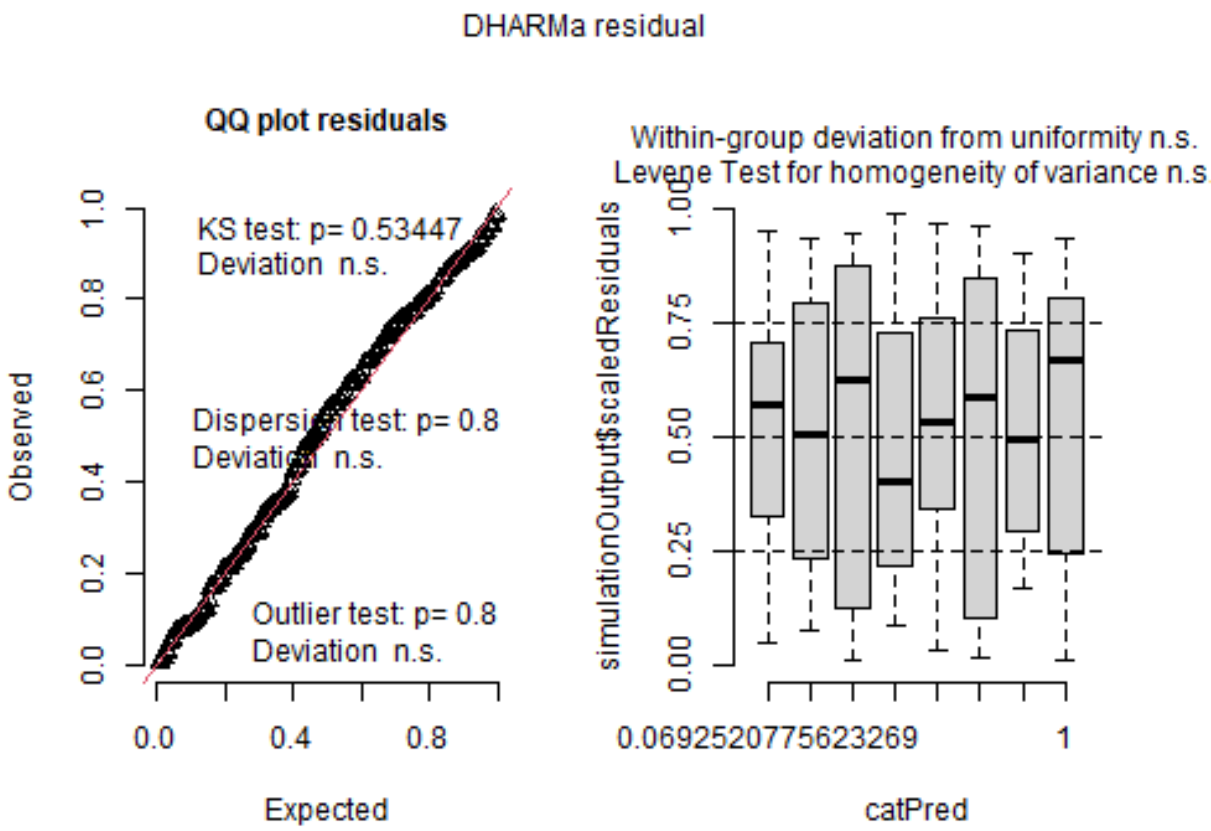
Model check

DHARMa

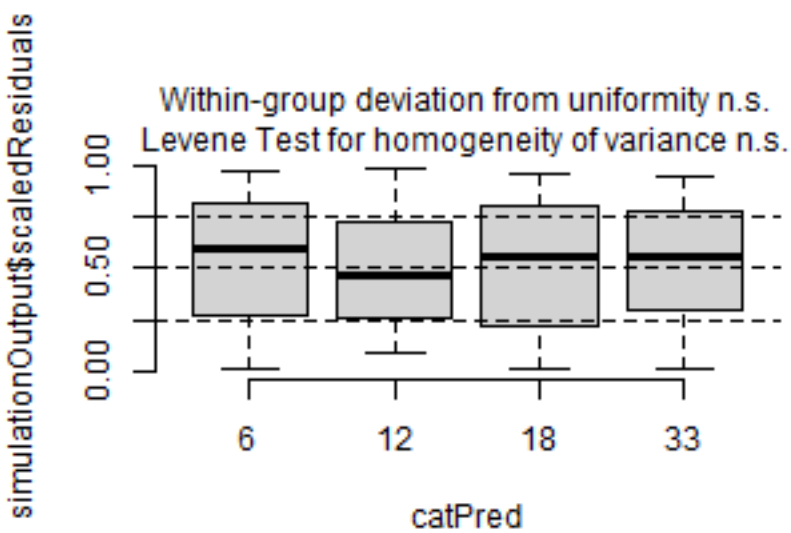
```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```



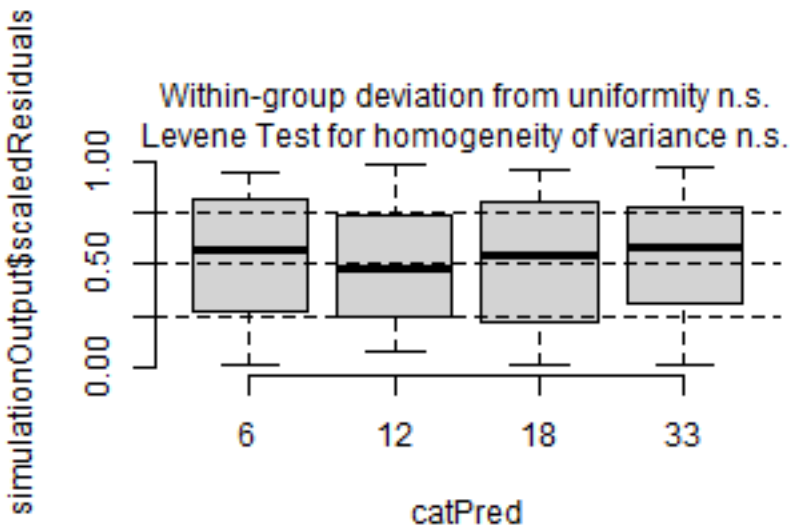

```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```



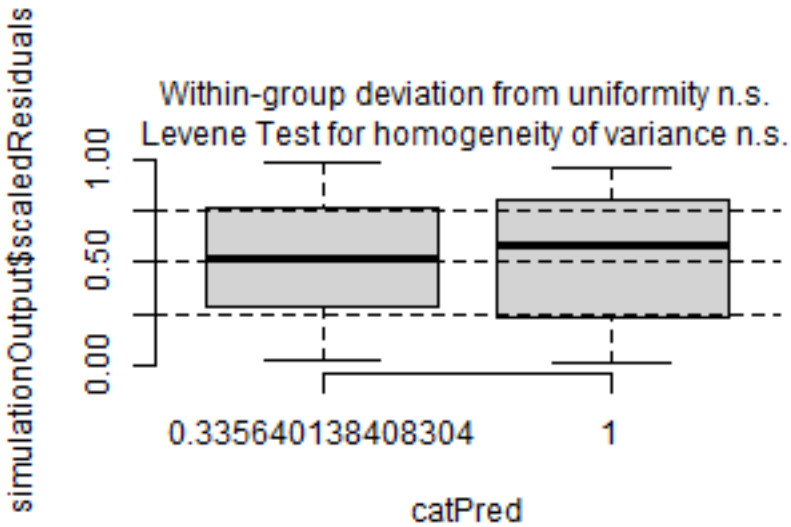

```
plotResiduals(simulation_output_1$scaledResiduals, factor(sites$seeded_pool))
```



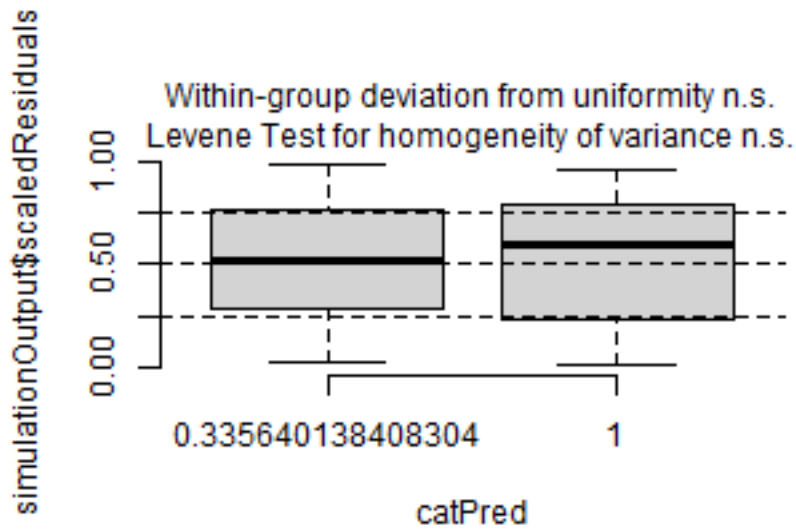

```
plotResiduals(simulation_output_2$scaledResiduals, factor(sites$seeded_pool))
```



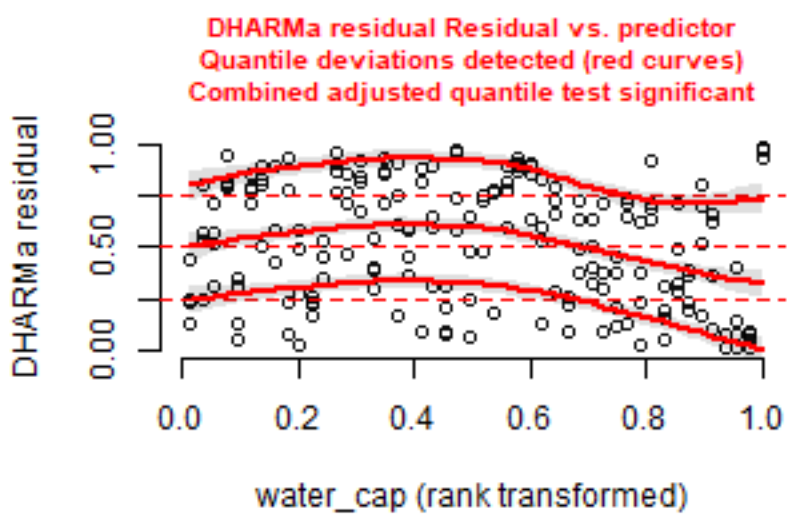

```
plotResiduals(simulation_output_1$scaledResiduals, sites$herbicide)
```



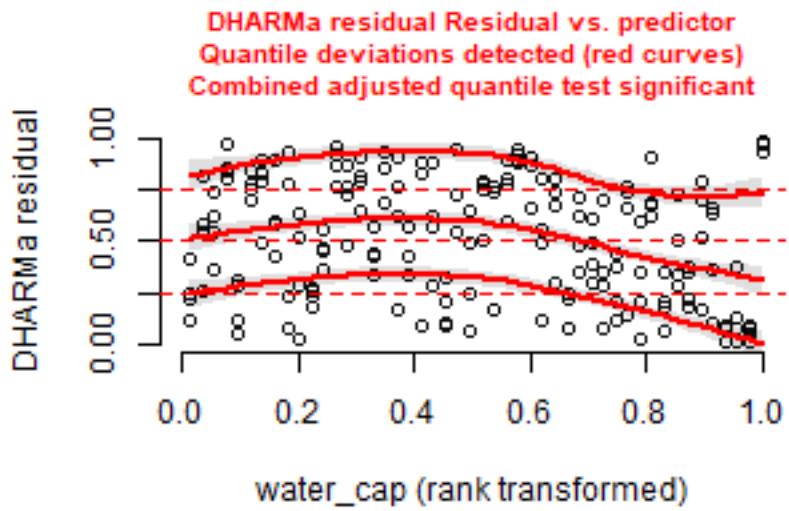

```
plotResiduals(simulation_output_2$scaledResiduals, sites$herbicide)
```




```
# plotResiduals(simulation_output_1$scaledResiduals, sites$seeding_time)
# plotResiduals(simulation_output_2$scaledResiduals, sites$seeding_time)
# plotResiduals(simulation_output_1$scaledResiduals, sites$year)
# plotResiduals(simulation_output_2$scaledResiduals, sites$year)
plotResiduals(simulation_output_1$scaledResiduals, sites$water_cap)
```

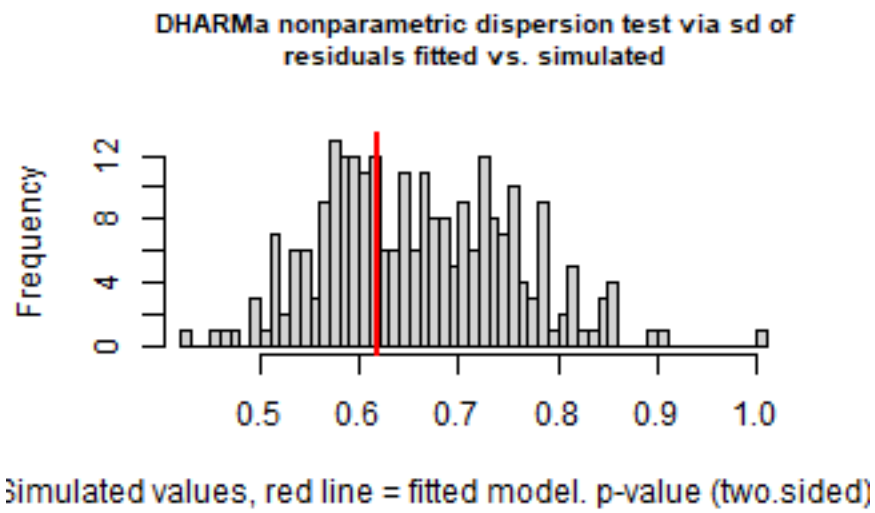



```
plotResiduals(simulation_output_2$scaledResiduals, sites$water_cap)
```



Test for overdispersion

```
testDispersion(simulation_output_1)
```




```
##
##  DHARMa nonparametric dispersion test via sd of residuals fitted vs.
##  simulated
##
## data:  simulationOutput
## dispersion = 0.93885, p-value = 0.8
## alternative hypothesis: two.sided
# testDispersion(simulation_output_2)
```

Check collinearity part 2 (Step 5)

Remove VIF > 3 or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
##               GVIF Df GVIF^(1/(2*Df))
```

```
## seeded_pool     1  3               1
## seeding_time    1  1               1
```

```
car::vif(m_2)
```

```
##                              GVIF Df GVIF^(1/(2*Df))
## seeded_pool              21.182058  3        1.663392
## seeding_time              1.094882  1        1.046366
## seeded_pool:seeding_time 22.075668  3        1.674888
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
## Warning: the null model is only correct if all the variables it uses are identical
## to those used in fitting the original model.
##                 R2m       R2c
## delta     0.3497473 0.3832592
## lognormal 0.3821614 0.4187791
## trigamma  0.3137853 0.3438514
MuMIn::r.squaredGLMM(m_2)
## Warning: the null model is only correct if all the variables it uses are identical
## to those used in fitting the original model.
##                 R2m       R2c
## delta     0.3378707 0.3719939
## lognormal 0.3698102 0.4071590
## trigamma  0.3025617 0.3331188
```

AICc

Use AICc and not AIC since ratio n/K < 40 Burnahm & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMIn::AICc(m_1, m_2) %>%
  arrange(AICc)
##     df     AICc
## m_1  6 730.6348
## m_2  9 736.4668
```

Predicted values

Summary table

```
car::Anova(m_2, type = 3)
```

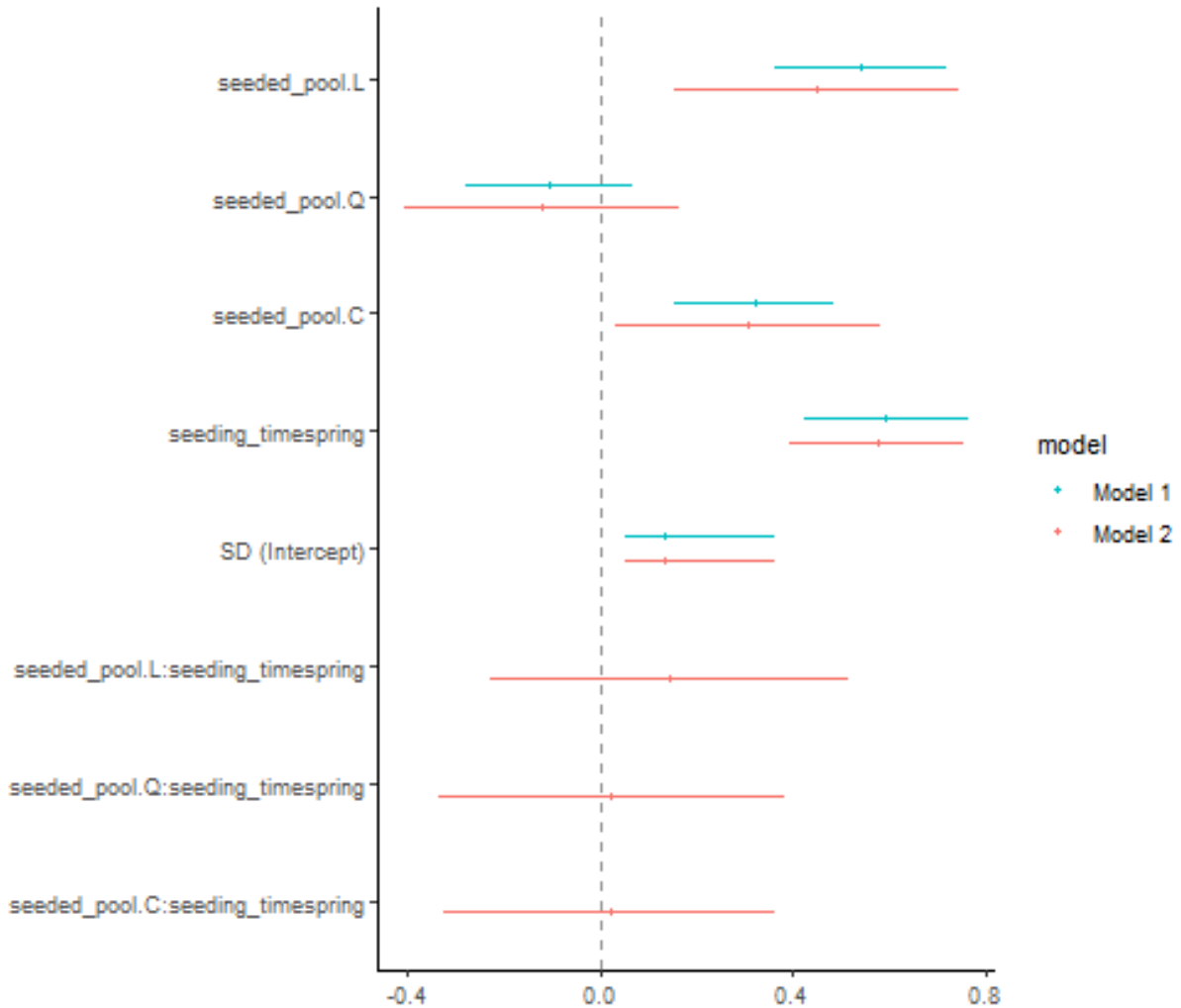
```
## Analysis of Deviance Table (Type III Wald chisquare tests)
##
## Response: y
##                            Chisq Df Pr(>Chisq)
## (Intercept)              50.8054  1  1.020e-12 ***
## seeded_pool              12.9976  3   0.004642 **
## seeding_time             39.6243  1  3.078e-10 ***
## seeded_pool:seeding_time  0.7027  3   0.872564
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
summary(m_2)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: poisson  ( log )
## Formula: y ~ seeded_pool * seeding_time + (1 | year)
##    Data: sites
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     735.5     764.8    -358.7     717.5       183
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.1437 -0.7093 -0.1005  0.6470  2.8989
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  year   (Intercept) 0.0182   0.1349
## Number of obs: 192, groups:  year, 4
##
## Fixed effects:
##                                  Estimate Std. Error z value Pr(>|z|)
## (Intercept)                       0.70826    0.09937   7.128 1.02e-12 ***
## seeded_pool.L                     0.44867    0.15054   2.980  0.00288 **
## seeded_pool.Q                    -0.12008    0.14538  -0.826  0.40881
## seeded_pool.C                     0.30633    0.14002   2.188  0.02869 *
## seeding_timespring                0.57414    0.09121   6.295 3.08e-10 ***
## seeded_pool.L:seeding_timespring  0.14231    0.18914   0.752  0.45181
## seeded_pool.Q:seeding_timespring  0.02290    0.18242   0.126  0.90011
## seeded_pool.C:seeding_timespring  0.02062    0.17544   0.118  0.90644
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) sdd_.L sdd_.Q sdd_.C sdng_t s_.L:_ s_.Q:_
## seeded_pl.L -0.168
## seeded_pl.Q  0.066 -0.270
## seeded_pl.C -0.114  0.054 -0.023
## sdng_tmsprn -0.583  0.183 -0.072  0.124
## sdd_pl.L:s_  0.133 -0.796  0.215 -0.043 -0.253
## sdd_pl.Q:s_ -0.053  0.215 -0.797  0.018  0.094 -0.291
## sdd_pl.C:s_  0.091 -0.043  0.018 -0.798 -0.160  0.056 -0.036
```

Forest plot

```r
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  theme_classic()
```



Effect sizes

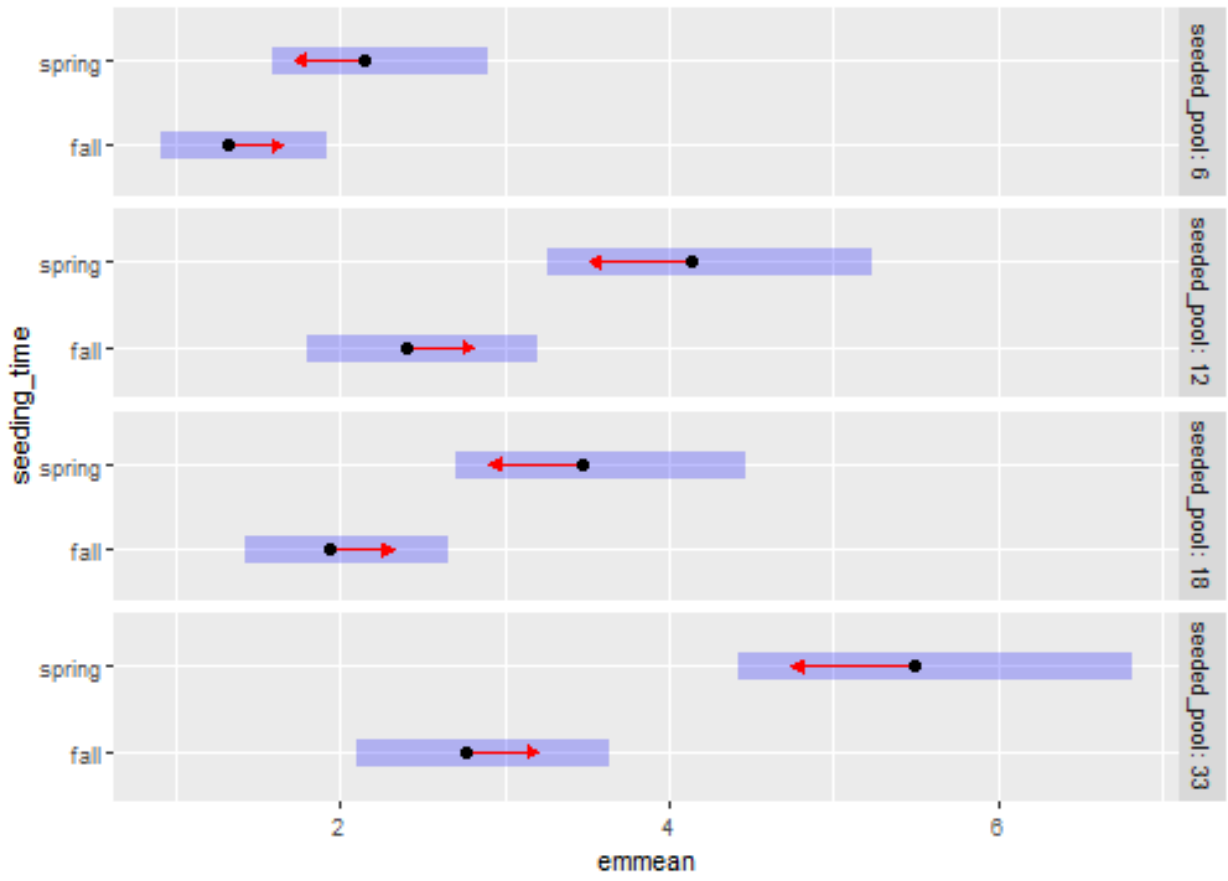
Effect sizes of chosen model just to get exact values of means etc. if necessary.

```
(emm <- emmeans(
  m_2,
  revpairwise ~ seeding_time | seeded_pool,
  type = "response"
  ))
```

```
## $emmeans
## seeded_pool = 6:
##  seeding_time rate    SE  df asymp.LCL asymp.UCL
##  fall         1.32 0.250 Inf     0.912      1.91
##  spring       2.15 0.331 Inf     1.587      2.91
##
## seeded_pool = 12:
##  seeding_time rate    SE  df asymp.LCL asymp.UCL
##  fall         2.40 0.354 Inf     1.793      3.20
##  spring       4.13 0.498 Inf     3.259      5.23
```

```
##
## seeded_pool = 18:
##  seeding_time rate    SE  df asymp.LCL asymp.UCL
##  fall         1.94 0.312 Inf     1.416      2.66
##  spring       3.47 0.445 Inf     2.697      4.46
##
## seeded_pool = 33:
##  seeding_time rate    SE  df asymp.LCL asymp.UCL
##  fall         2.77 0.386 Inf     2.105      3.64
##  spring       5.49 0.604 Inf     4.427      6.81
##
## Confidence level used: 0.95
## Intervals are back-transformed from the log scale
##
## $contrasts
## seeded_pool = 6:
##  contrast      ratio    SE  df null z.ratio p.value
##  spring / fall  1.63 0.365 Inf    1   2.163  0.0305
##
## seeded_pool = 12:
##  contrast      ratio    SE  df null z.ratio p.value
##  spring / fall  1.72 0.284 Inf    1   3.303  0.0010
##
## seeded_pool = 18:
##  contrast      ratio    SE  df null z.ratio p.value
##  spring / fall  1.79 0.325 Inf    1   3.191  0.0014
##
## seeded_pool = 33:
##  contrast      ratio    SE  df null z.ratio p.value
##  spring / fall  1.99 0.297 Inf    1   4.581  <.0001
##
## Tests are performed on the log scale
```

```r
plot(emm, comparison = TRUE)
```



Session info

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats     graphics  grDevices utils     datasets  methods   base
##
## other attached packages:
##  [1] emmeans_1.11.1   DHARMa_0.4.7     glmmTMB_1.1.12   lme4_1.1-37
##  [5] Matrix_1.7-3     patchwork_1.3.1  ggbeeswarm_0.7.2 lubridate_1.9.4
##  [9] forcats_1.0.0    stringr_1.5.1    dplyr_1.1.4      purrr_1.1.0
```

```
## [13] readr_2.1.5      tidyr_1.3.1      tibble_3.3.0     ggplot2_3.5.2
## [17] tidyverse_2.0.0  here_1.0.1
##
## loaded via a namespace (and not attached):
##  [1] mnormt_2.1.1           Rdpack_2.6.4           gridExtra_2.3
##  [4] sandwich_3.1-1         rlang_1.1.6            magrittr_2.0.3
##  [7] compiler_4.5.1         mgcv_1.9-3             vctrs_0.6.5
## [10] quadprog_1.5-8         pkgconfig_2.0.3        crayon_1.5.3
## [13] fastmap_1.2.0          backports_1.5.0        labeling_0.4.3
## [16] pbivnorm_0.6.0         utf8_1.2.6             ggstance_0.3.7
## [19] promises_1.3.3         rmarkdown_2.29         tzdb_0.5.0
## [22] nloptr_2.2.1           bit_4.6.0              xfun_0.52
## [25] later_1.4.2            lavaan_0.6-19          parallel_4.5.1
## [28] R6_2.6.1               gap.datasets_0.0.6     stringi_1.8.7
## [31] qgam_2.0.0             RColorBrewer_1.1-3     car_3.1-3
## [34] boot_1.3-31            numDeriv_2016.8-1.1    estimability_1.5.1
## [37] Rcpp_1.1.0             iterators_1.0.14       knitr_1.50
## [40] zoo_1.8-14             parameters_0.27.0      httpuv_1.6.16
## [43] splines_4.5.1          timechange_0.3.0       tidyselect_1.2.1
## [46] rstudioapi_0.17.1      abind_1.4-8            yaml_2.3.10
## [49] MuMIn_1.48.11          doParallel_1.0.17      TMB_1.9.17
## [52] codetools_0.2-20       nonnest2_0.5-8         lattice_0.22-7
## [55] plyr_1.8.9             shiny_1.11.1           withr_3.0.2
## [58] bayestestR_0.16.1      coda_0.19-4.1          evaluate_1.0.4
## [61] marginaleffects_0.28.0 CompQuadForm_1.4.4     pillar_1.11.0
## [64] gap_1.6                carData_3.0-5          foreach_1.5.2
## [67] stats4_4.5.1           reformulas_0.4.1       insight_1.3.1
## [70] generics_0.1.4         vroom_1.6.5            rprojroot_2.0.4
## [73] hms_1.1.3              scales_1.4.0           minqa_1.2.8
## [76] xtable_1.8-4           glue_1.8.0             tools_4.5.1
## [79] data.table_1.17.8      mvtnorm_1.3-3          grid_4.5.1
## [82] rbibutils_2.3          datawizard_1.1.0       nlme_3.1-168
## [85] Rmisc_1.5.1            performance_0.15.0     beeswarm_0.4.0
## [88] vipor_0.4.7            Formula_1.2-5          cli_3.6.5
## [91] gtable_0.3.6           digest_0.6.37          farver_2.1.2
## [94] htmltools_0.5.8.1      lifecycle_1.0.4        mime_0.13
## [97] bit64_4.6.0-1          dotwhisker_0.8.4       MASS_7.3-65
```